

Westchester County Data Platform

Reporting Strategy & Output Catalog

Arcanum Research Initiative

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Contents

1 Overview

This document catalogs all outputs from the Westchester County Data Platform project and provides guidance on accessing, using, and maintaining the various deliverables. The platform follows Druck organizational standards with a two-folder structure (Output/ and Technical/) ensuring clear separation between user-facing deliverables and implementation details.

2 Output Structure

2.1 Directory Organization

```
Westchester/
  Output/                                # User-facing deliverables
    Data/
      Results/                          # Processed Excel files
      Source/                          # Original data documentation
      Robin/                          # Robin database integration
    PDFs/                              # LaTeX-generated reports
    Charts/                          # Static visualizations
    Documentation/                    # Analysis guides
  Technical/                          # Implementation files
    src/                              # Source code
    data/                            # Raw and processed data
    notebooks/                      # Jupyter analysis
    docs/                          # LaTeX sources
    scripts/                        # Automation scripts
```

3 Excel Data Files

3.1 Druck Compliance Standards

All Excel files follow mandatory Druck standards:

- **ONE sheet per file** (no multi-tab workbooks)
- Machine-readable column names
- Professional black & white formatting
- Timestamped filenames: [YYYY.MM.DD] description.xlsx

3.2 Excel File Catalog

File Name	Description
[2025.10.13] westchester_county_demographics.xlsx	County-level demographic statistics from 2022 ACS 5-Year Estimates. Includes population, income, housing, and employment data.

File Name	Description
[2025.10.13] westchester_census_tracts.xlsx	Demographic data for all 241 census tracts in Westchester County. Tract-level population, income, and housing metrics.
[2025.10.13] westchester_metro_north_stations.xlsx	Directory of 56 Metro-North Railroad stations in Westchester County with coordinates and accessibility information.
[2025.10.13] westchester_transit_accessibility.xlsx	Transit accessibility analysis including station counts, ADA compliance rates, and coverage metrics.

Table 1: Excel Data Files Inventory

3.3 How to Use Excel Files

1. Navigate to `Output/Data/Results/`
2. Open files in Excel, Google Sheets, or any spreadsheet application
3. Each file contains exactly ONE sheet with all data
4. Column headers are self-explanatory and machine-readable
5. All files are timestamped for version control

4 PDF Reports

4.1 Report Catalog

Report	Content
westchester_methodology_report.pdf	Comprehensive documentation of data sources, processing methods, analytical framework, and technical implementation details.
westchester_executive_summary.pdf	High-level overview of key findings, deliverables, and applications. Non-technical summary for stakeholders and decision-makers.
westchester_reporting_strategy.pdf	This document. Complete catalog of all outputs with usage instructions and maintenance guidelines.

Table 2: PDF Reports Inventory

4.2 Compiling LaTeX Reports

To regenerate PDF reports from source:

```
cd Technical/docs/
pdflatex westchester_methodology_report.tex
pdflatex westchester_executive_summary.tex
```

```
pdflatex westchester_reporting_strategy.tex
```

```
# Move PDFs to Output/PDFs/
mv *.pdf ../../Output/PDFs/
```

5 Web Platform

5.1 Starting the Platform

Quick Start:

```
# Backend API
cd Technical/src/api
python main.py
# Runs on http://localhost:8000
```

```
# Frontend Application
cd Technical/src/frontend
npm run dev
# Runs on http://localhost:5173
```

5.2 Dashboard Catalog

Dashboard	URL & Features
Overview Dashboard	/dashboards/overview High-level county metrics, data availability status, interactive map
Demographics Dashboard	/dashboards/demographics Population charts, income distribution, race/ethnicity breakdown, housing statistics
Transit Dashboard	/dashboards/transit Station directory, accessibility analysis, coverage maps, 1-mile radius visualization
Property Tax Dashboard	/dashboards/property-tax Assessment trends, tax rates, geographic distribution by municipality
Budget Dashboard	/dashboards/budget County spending by department, multi-year budget trends, spending priorities
Municipal Services Dashboard	/dashboards/services Police, fire, library coverage; emergency response times; facility inventory
Municipality Comparison	/dashboards/comparison Side-by-side comparison of demographics, income, and housing across municipalities

Dashboard	URL & Features
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Table 3: Web Dashboard Catalog

6 API Endpoints

6.1 API Documentation

Interactive API documentation available at: <http://localhost:8000/docs>

6.2 Endpoint Catalog

Endpoint	Description
GET /	API root with welcome message and status
GET /api/stats	Platform statistics and data availability
GET /api/municipalities	List of Westchester County municipalities
GET /api/transit/stations	Metro-North station data with coordinates
GET /api/demographics/county	County-level demographics (2022 ACS)
GET /api/demographics/tracts	Census tract demographics (241 tracts)
GET /api/demographics/municipalities	Municipality-level demographics

Table 4: API Endpoints

7 Data Update Schedule

7.1 Recommended Update Frequency

Data Source	Update Frequency	Command
Census Demographics	Annually	<code>python scripts/download_all_data.py</code>
Metro-North GTFS	Monthly	<code>python scripts/download_all_data.py</code>
NY State Data	Quarterly	<code>python scripts/download_all_data.py</code>
Excel Files	After data updates	<code>python src/processors/generate_all_excel.py</code>

7.2 Update Process

1. Run data download script
2. Verify data quality
3. Regenerate Excel files

4. Validate Druck compliance
5. Update documentation if needed
6. Restart web platform

8 Quality Assurance

8.1 Validation Procedures

Excel Validation:

```
python Technical/scripts/validate_excel.py
```

Checks for ONE sheet per file, validates column names, verifies formatting

API Testing:

```
# Access API documentation
```

```
http://localhost:8000/docs
```

```
# Test each endpoint interactively
```

8.2 Compliance Checklist

All Excel files have exactly ONE sheet

Filenames include timestamp [YYYY.MM.DD]

Column names are machine-readable

All PDFs generated from LaTeX sources

API endpoints return valid JSON

Frontend dashboards load without errors

Documentation is up to date

9 User Support

9.1 Documentation Resources

- **Project README:** Projects/Westchester/README.md
- **Technical Guide:** Projects/Westchester/Technical/README.md
- **Methodology Report:** Output/PDFs/westchester_methodology_report.pdf
- **API Docs:** <http://localhost:8000/docs>

9.2 Common Tasks

Accessing Data:

- Excel files: `Output/Data/Results/`
- PDF reports: `Output/PDFs/`
- Raw data: `Technical/data/raw/`

Running Analysis:

- Jupyter notebooks: `Technical/notebooks/`
- Python scripts: `Technical/scripts/`
- API queries: Use interactive docs at `/docs`

10 Maintenance

10.1 Regular Maintenance Tasks

1. **Monthly:** Update Metro-North GTFS data
2. **Quarterly:** Refresh NY State Open Data
3. **Annually:** Update Census ACS data when new year is released
4. **As Needed:** Add new data sources, expand dashboards

10.2 Version Control

- All Excel files are timestamped
- Git repository tracks all code changes
- Documentation updated with each major release

11 Contact Information

For questions, issues, or requests for additional analysis:

- **Project:** Westchester County Data Platform
- **Organization:** Arcanum Research Initiative
- **Documentation:** See `HANDOFF_DOCUMENTATION.md`